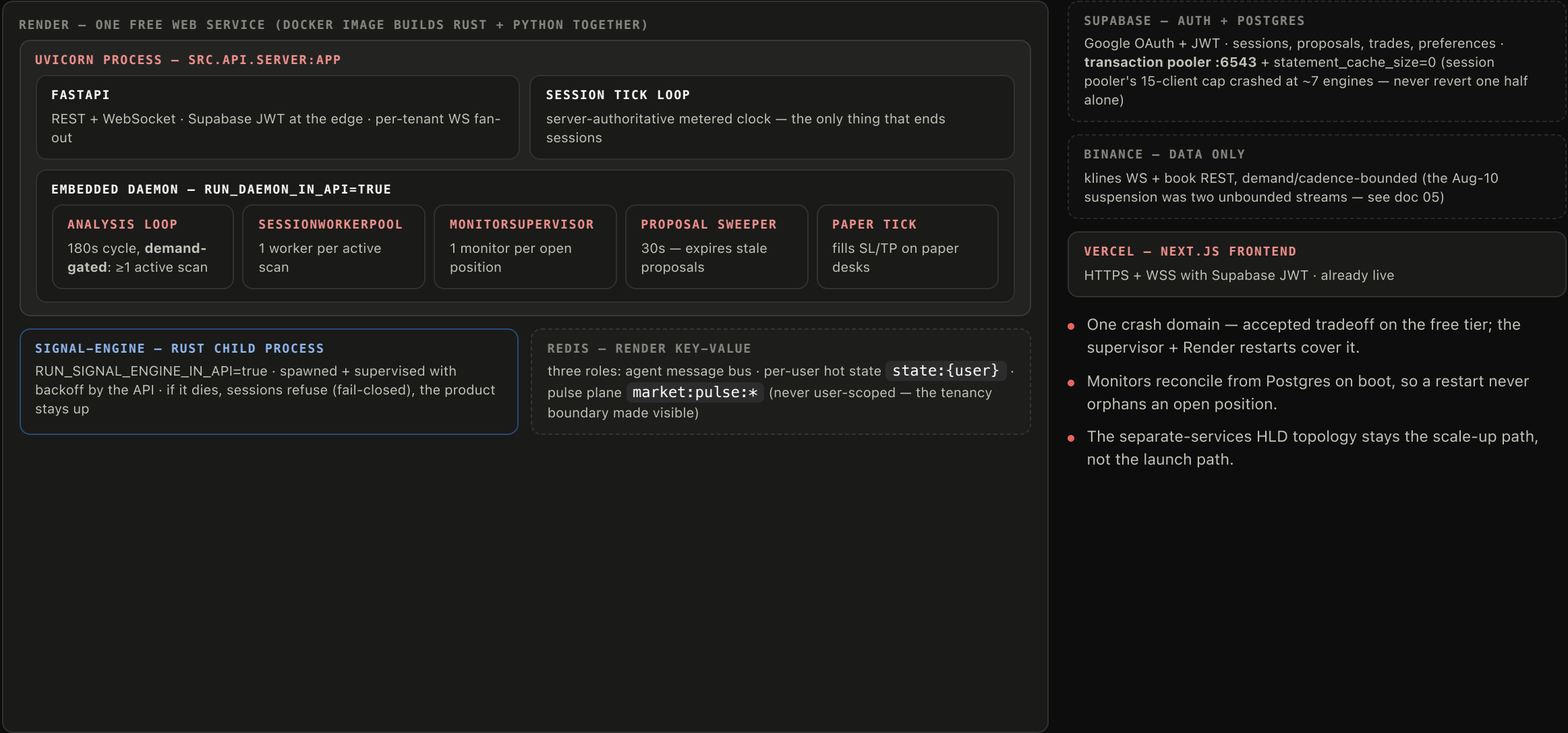


System *design*

The architecture that already runs — one process, three planes, demand-gated compute — plus the exact deltas that turn it into the 2-slot free tier: a slot quota, a trade-focus lock, a hot-window surface, and zero always-on waste.

Doc	Date	Builds on
03 of 06 — System design	2026-08-24	docs/ARCHITECTURE.md (verified 2026-08-11)

One process is the whole backend — by design, for the free tier



- One crash domain — accepted tradeoff on the free tier; the supervisor + Render restarts cover it.
- Monitors reconcile from Postgres on boot, so a restart never orphans an open position.
- The separate-services HLD topology stays the scale-up path, not the launch path.

Shared facts, per-user judgment, unmetered protection

PLANE 1 · SHARED – USER-BLIND, O(1) IN USERS

Rust engine: ingest (klines WS · book REST) → feature store → 5 vetoes → 6 weighted factors → **regime + tradability 0–100** → **market:pulse:SYMBOL**, refreshed ≤5s, TTL 120s.

SHARED LLM ANALYSIS

Data → Analysis → Strategy agents produce a **candidate-setup pool per symbol** (SharedSetupCache) — computed once per cycle for everyone, demand-gated.

NEW · HOT-WINDOW SURFACE

market:pulse:global — cross-symbol "hot / quiet" now-cast + slots-aware nudge. Every pulse also appends to **pulse_snapshots** so the score stays falsifiable.

The pulse carries facts and scores only — never a direction, entry, or stop. Regime-direction was refuted out-of-sample; a contract test enforces the ban.

PLANE 2 · PER-USER – O(N) IN ACTIVE SCANS

SessionWorker (metered, 1 per scan): reads prefs + fresh pulses — **fail-closed at 15s staleness** — then **analyze_fn**: pulse gate → style filter → **per-user synthesis** (LLM selects the candidate that fits *this* trader and writes the thesis in their terms — it never invents a price) → deterministic sizing vs the user's own risk rules.

PROPOSAL (POSTGRES)

entry / stop / targets / size / leverage / R:R / rationale / shelf life. Clock **pauses** while the user decides. Approval **re-validates at the live price** — drift >0.5% refuses rather than fills worse than agreed.

Per-user LLM budget: **analyze_fn** returns real usage; the cap guards a moving number, fail-closed.

FIX BEFORE ENABLING THE PLANE

Confirmed bug: the pipeline calls **.get()** on **Pulse** dataclasses — enabling **SIGNAL_PLANE_ENABLED** today fails **every metered cycle** while the user's clock drains. Normalize at the boundary (as **monitor_worker** already does) + fold pulse liveness into the capacity check so dead-plane time is refunded.

PLANE 3 · EXECUTION + WATCH

Per-user engine (paper or live) books the bracket. Two money gates fail closed: **ENABLE_EXECUTION** + **LIVE_TRADING_CONFIRMED**.

POSITIONMONITOR – UNMETERED

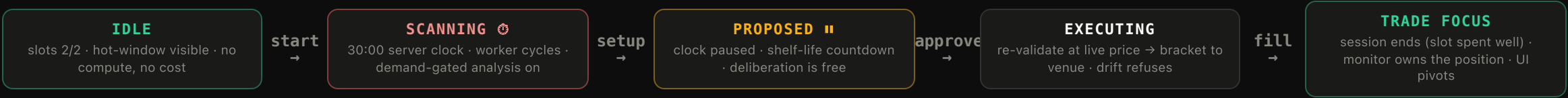
1 per open position. Time stop · profit give-back · **conditions-turned** (pulse drift) → reduce-only exit when the thesis breaks. Outlives the session, restarts, deploys; supervisor reconciles on boot.

BOOKKEEPING

Fill → trade entry row (memory loop feeds) → portfolio + history update → session slot state advances (free tier: focus lock until flat).

The session state machine, extended for slots and trade focus

The metered state machine exists and is well-tested. Two mechanics change for the new product: quota becomes **2 discrete slots/day** (not a minutes pool), and a **focus lock** holds new slots while a free-tier position is open.



SLOT ACCOUNTING (REPLACES MINUTES POOL)

- Today: one pooled 1800s/day quota, **identical for every tier**, unlimited session count (30 one-minute sessions is legal). Replaced by per-plan policy: `slots_per_day × slot_minutes`, fixed grant per slot.
- Slot consumed at **start**, never refunded for boredom; refunded only for **our** failures (the capacity-loss refund path already exists).
- Client-supplied `quota_seconds` is gone — tier config is the only quota source.

THE FOCUS LOCK (FREE TIER)

- Invariant: `open_positions ≥ 1 ⇒ session_start refused` with an honest reason + the monitor's live status.
- Proposal flow already ends the session on `trade_opened` — the lock closes the loop by holding the *next* start until flat.
- Paid tiers raise the concurrency cap instead of dropping the rule — same code path, different limit.

REJECT / EXPIRE / END PATHS

- Reject or shelf-life expiry → back to SCANNING, clock resumes — covered today.
- Quota spent or user ends → ENDED; open positions keep their monitors — quota never stops a monitor.
- Expiry sweeper ends sessions of disconnected users — ticks must not depend on an open WebSocket.

Demand-gated everything: idle costs zero, sessions cost minutes

The founder's instinct becomes the architecture's law: no stream or loop runs without a consumer. The audit enumerated all 25 loops (doc 01); the "always-running 300s loop" (EventCoordinator) was dead code — already attic'd. One blocker remains in live code.

WORKLOAD	CADENCE	TARGET BEHAVIOR	BANDWIDTH (EXTERNAL)	STATUS
Daemon DataAgent streams — depth20@100ms + markprice@1s + ticker + 5 kline TFs, per symbol	24/7	Gate to scan demand (same signal as analysis) + drop depth/ticker to REST snapshots at cycle cadence	~2.2 GB/day today → ~0 idle	R0 BLOCKER
Signal engine ingest (child process)	5s score	REST-poll closed bars per TF boundary (bootstrap already does REST) + 5s batched book poll; bookTicker stream already removed	~7.4 GB/mo as WS → ~1 GB/mo as REST	R1 · K LINES→REST
Shared LLM analysis cycle	180s	≥1 active scan or explicit keep-warm — zero scans = zero LLM calls (uncommitted diff completes this)	LLM only	DEMAND-GATED
SessionWorkers · PositionMonitors	180s / 15s	1 per active session · 1 per open position; none exist otherwise	internal	DEMAND-GATED
Dashboard market feed	stream	refcounted to live WS viewers + 30s linger (commit 9edada0) — but client-requested extra channels are add-only: refcount them too	~9 MB per viewed hour	FIXED +R0 REFCOUNT
Idle Binance socket	~90s	skip staleness check while zero streams subscribed — today it reconnect-churns ~960×/day	~6 MB/day	R0 · SMALL FIX
Sweeper · session tick · paper tick · heartbeats	30s/5s/3s	always-on but internal-only (Redis/Postgres); idle no-ops against empty sets	0	KEEP
Keepalive cron (GitHub Actions)	10 min	the one deliberate always-on — keeps Render awake so monitors survive	~0.2 MB/day	KEEP

Every stream, loop, and LLM call must name its consumer. Subscribing at process startup is forbidden; tie it to a live session, a live viewer, or an open position — with a linger, then

Schema deltas: small, additive, forward-only

CHANGES FOR THE SLOT MODEL + TIERS	
DELTA	DETAIL
tier config	plans move from env vars to a <code>plans</code> config (free / lite / pro / pro_max): slots_per_day, slot_minutes, max_concurrent_trades, channels, async flag — free tier hardcoded live, others dormant
sessions	slot semantics: <code>slot_index</code> per day; daily reset job keys off user's tier; keeps existing state machine columns
user_preferences	already has capital, risk %, goal fields — onboarding finally writes them; add <code>onboarding_completed_step</code> for resume
pulse_snapshots	gains its writer (engine appends every publish) — the calibration log the quant desk requires
trades	entry rows written on every book (memory loop) — exit updates must never hit "not found"

- DEBTS ACKNOWLEDGED, SCHEDULED, NOT BLOCKING FREE LAUNCH

 - **Money as floats** everywhere except `llm_cost_micros` — one wholesale integer-minor-units + currency migration lands **before any real-money path opens** (roadmap gate, doc 06). Paper-only launch tolerates floats; real money never will.
 - **RLS defense-in-depth** on the five new per-user tables — backend bypasses RLS, so app-layer `user_id` filters remain the only guard; RLS is the second lock.
 - **Alembic vs checkfirst collision** — stamp the production DB once, then forward-only migrations in the deploy step.
- TENANCY INVARIANTS (TESTED, KEEP ENFORCING)

 - Every per-user table carries an indexed `user_id`; `pulse_snapshots` must never gain one — schema-parity test enforces both.
 - Every new query filters `user_id` in app code; a missing filter is a launch blocker, not a bug.
 - Per-user Redis keys stay namespaced `state:{user_id}:*`; pulse keys stay global.

API deltas: four new reads, one changed start, nothing else moves

NEW / CHANGED ENDPOINTS

ENDPOINT	CHANGE
GET /api/pulse	new — per-symbol regime + tradability + global hot/quiet, authenticated, cached 5s server-side; the dashboard's banner + strip read this
GET /api/session/slots	new — slots used/remaining today, focus-lock state + reason, next reset time
POST /api/session/start	changed — no client <code>quota_seconds</code> ; consumes a slot server-side; 409 with monitor status when focus-locked; 503 (no row, no clock) when analysis capacity is down
GET /api/monitor/:position	new — the monitor's verdict log for the trade-focus surface
GET /api/session/recap/:id	new — setups seen / action taken / outcome, powers the recap card

HARDENING THAT RIDES ALONG (FROM THE AUDIT, DOC 01)

- `require_admin` fail-closed; anonymous fail-open mode impossible in prod config
- WS auth moves from `?token=` query param to first-message auth (JWTs out of proxy logs)
- The static `NEXT_PUBLIC_API_TOKEN` tier dies — Supabase JWT is the only client credential
- Destructive one-off scripts leave the image (attic'd + .dockerignore)

CONTRACT RULES

- Every tenant endpoint requires the JWT and scopes by `user_id` — `/api/setups` 's unauthenticated broadcast model is retired with the v2 daemon path.
- Pulse responses carry `as_of` + schema version; consumers reject stale or unknown-major — same contract the workers obey.
- Errors are honest: capacity down = 503 + "engine unavailable", never a silent empty list.